

Fraud-Resistant Meal Voucher Redemption and Real-Time Operational Analytics for Conference Management: Design and Evaluation of ConferenceCheck Mobile

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September 2026

Repository: <https://github.com/ndashi-chisanga6/ConferenceCheck-Mobile-Analytics-Meals-and-Notifications> Validated state: tag milestone-10-evaluation

Declaration

I declare that this report is my own work. It has not been submitted, in whole or in part, for any other award at this or any other institution, and every source I have drawn on is acknowledged in the reference list.

The use of artificial intelligence tools during this project is disclosed in **Appendix A**, which records what each category of tool was used for, what was provided to it, and the limits placed on its use by Section 9 of the approved project proposal. All evaluation data in this study is seeded and synthetic: attendees, meal vouchers and sessions in the simulated-event exercise are generated by `tools/seed-evaluation-event.php` and describe no real conference or real person. No respondent or attendee data belonging to any third party was uploaded to any external tool at any point.

Signed: Date:

Conventions used in this report

Latencies are quoted in milliseconds to one decimal place in tables and rounded to the nearest whole millisecond in prose. Counts and rejection rates are exact throughout. Timings are wall-clock seconds or milliseconds for a single named run on PHP 8.5.8 with PostgreSQL 18, and the run is always named. Where a number also appears in `docs/validation_report.md`, the two

are the same measurement and that file is the reproducible source; this report does not re-derive figures independently of it.

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Abstract

On-site conference operations, meal service, session seating and attendee communication, are still widely managed with paper registers and printed vouchers, processes that scale poorly and are vulnerable to duplicate-collection fraud. This report presents the design, implementation and evaluation of ConferenceCheck Mobile: Analytics, Meals and Notifications, component 4.3b of a two-part conference management companion application whose sibling component, 4.3a, delivers attendee check-in capture and offline synchronisation separately. This project consumes check-in data through a shared REST API contract and contributes four capabilities that carry equal weight in its evaluation: a QR-code based meal voucher protocol in which single-use redemption is enforced structurally, through database-level row locking and an append-only redemption relation with a per-voucher uniqueness constraint, rather than through application-level checks; a near-real-time analytics dashboard with a bounded 30-second staleness guarantee; session attendance tracking with configurable capacity thresholds and automatic overcrowding alerts; and dual-channel attendee notifications, combining Firebase Cloud Messaging push with an authenticated in-app feed carrying per-recipient delivery records. An offline scan queue, whose safe replay follows from the server's idempotent scan semantics, extends the design to intermittent venue connectivity. The system was evaluated against a seeded 500-attendee event with three meal categories and a voucher per attendee per category, ten sessions and scripted concurrent-fraud scenarios, driven through the live API rather than through unit-level mocks. Every criterion in the proposal's Table 3 was met: 100% duplicate-redemption rejection across 100 sequential and 20 concurrent duplicate submissions, with every concurrency race resolving to exactly one success; all three categories redeemable independently for every one of twenty attendees holding a voucher in each; median voucher-scan latency of 187.5 ms (95th percentile 286.7 ms); dashboard staleness bounded at approximately 30.2 seconds; and byte-exact CSV exports against database ground truth for both attendance and meal redemptions. A live, human-staffed manual baseline was not run, and this is stated as a limitation rather than approximated with a fabricated figure; the qualitative argument instead rests on the system's measured 188 ms scan-to-decision time and the fact that a paper voucher has no server-side check to fail. The results indicate that placing anti-fraud guarantees in the database's constraint machinery, rather than in application code, eliminates time-of-check-to-time-of-use races by construction while keeping scan-point latency well within interactive bounds.

1. Introduction

Modern conferences concentrate hundreds of attendees, multiple parallel sessions and time-boxed catering windows into a short operational period. The quality of the attendee experience depends on how efficiently organisers can verify entitlement, observe live operational state,

and communicate changes as they happen. Traditional approaches rely on fragmented, largely manual tools: printed meal tickets, paper attendance registers and public-address announcements. These methods have three structural weaknesses. Paper vouchers are trivially duplicated or reused, and manual meal lines create congestion at service points. Organisers have no consolidated, real-time view of attendance or consumption, so operational decisions are made on stale or anecdotal information. Attendee communication is broadcast-only and unverifiable, with no record of who received a given update.

Quick Response (QR) codes provide a mature, standardised mechanism for machine-readable identification [1], and QR-based attendance systems have been shown to reduce both processing time and recording error relative to manual registers [3]. Cloud push-messaging services such as Firebase Cloud Messaging provide reliable, low-latency delivery of updates to mobile devices at no marginal cost [6], and modern cross-platform toolkits such as Flutter allow a single codebase to deliver the required scanning, charting and notification functionality [15]. This project applies these technologies to conference operations as component 4.3b of the ConferenceCheck capstone programme. A sibling component, 4.3a, handles the core attendee check-in capture and offline synchronisation engine separately; this project does not rebuild those features, and instead consumes check-in data through a shared REST API contract and focuses on what happens after check-in.

1.1 Problem statement

Conference organisers lack integrated, trustworthy tools for the operational tasks that follow attendee registration and check-in. Meal distribution based on printed vouchers or name lists cannot enforce single use; duplicate collection is difficult to detect at the point of service, and reconciliation after the event is manual and unreliable. Attendance and check-in data, even when captured digitally, is not surfaced to organisers in real time, so decisions about catering quantities, room allocation and staffing are made on guesswork. Session rooms have physical capacity limits, but organisers have no mechanism that tracks per-session attendance as it happens or warns them before rooms overcrowd. Communication with attendees is one-directional and unlogged, with no record of which attendees were reachable for a given schedule change. The consequence is operational inefficiency, measurable fraud exposure in catering budgets, safety risk from overcrowded rooms, and a degraded attendee experience.

1.2 Aim

To design, implement and evaluate the analytics, meal management, session capacity, notification and reporting component of the ConferenceCheck Mobile application, providing conference organisers with real-time operational insight, fraud-resistant meal voucher redemption and auditable attendee communication.

1.3 Objectives

1. To design and implement a real-time analytics dashboard showing check-in progress, check-in rates and trends for event organisers.

2. To build a QR-code based meal voucher scanning and redemption system that enforces single-use redemption across multiple meal categories.
3. To implement session attendance tracking with capacity management and overcrowding alerts.
4. To build a push notification system for attendee communications and schedule updates, with per-recipient delivery records.
5. To implement reporting and data export (CSV) functionality for event organisers.
6. To produce comprehensive technical documentation and a publication-ready paper.

1.4 Scope

The project covers Flutter mobile modules for the analytics dashboard, meal voucher generation and scanning, session attendance and capacity alerts, a notification inbox and report access, together with the corresponding server-side business logic on the CMS v4 Laravel backend. It integrates with component 4.3a's core check-in module through a shared REST API contract, consuming its check-in data as a read-only input to analytics; check-in scanning, attendee search and the offline synchronisation engine are component 4.3a's remit and are out of scope here. Evaluation is performed on seeded, realistic event datasets including concurrent-scan fraud scenarios, described in full in Section 5. Payment processing, external ticketing integration and a full web-based administration panel are out of scope for both components.

1.5 Contributions

This work makes four contributions, each carrying equal weight in the evaluation that follows. The first is a redemption protocol in which the single-use guarantee is carried by the database's constraint machinery: a unique constraint on an append-only redemption relation, enforced inside a transaction with a row lock on the voucher, rather than by an application-level check, which eliminates time-of-check-to-time-of-use races by construction rather than by defensive coding. The second is a near-real-time analytics dashboard whose staleness bound is a stated design parameter (a 30-second poll) rather than an incidental property, chosen explicitly over introducing a WebSocket layer within the project's scope. The third is session capacity tracking whose overcrowding alerts are dispatched automatically on a threshold transition, converting a safety problem into an alert delivered before a room reaches capacity rather than a discovery made after. The fourth is a dual-channel notification design, push plus an authenticated in-app inbox with per-recipient delivery records, that treats Firebase Cloud Messaging's best-effort delivery model as a known property to be compensated for rather than assumed away, together with an offline scan queue whose replay safety is a direct corollary of the same idempotent server design that makes the redemption protocol safe.

2. Literature Review

The QR Code symbology is standardised in ISO/IEC 18004 [1], which specifies its data capacity and, critically for verification applications, its four selectable error-correction levels, allowing codes to remain machine-readable when partially damaged or poorly displayed on low-

brightness screens. Tiwari [2] surveys the structural properties that drove QR adoption over one-dimensional barcodes in ticketing and access-control contexts: high capacity in a small print area, omnidirectional scanning, and decode robustness on commodity camera hardware. A key design implication, adopted in this project, is that the QR payload should be an opaque server-issued token rather than semantically meaningful data, so that possession of the printed code confers no information and validation authority remains entirely server-side.

Masalha and Hirzallah [3] evaluated a QR-based attendance system in a university setting and found substantial reductions in per-person processing time and in recording errors relative to manual registers, but also identified the central weakness of naive QR attendance: codes can be photographed and shared, so any anti-fraud property must come from what the server does with a scan, not from the code itself. This finding directly motivates the server-side single-use redemption design evaluated in this report, in which the database, not the scanning client, is the arbiter of whether a voucher has already been consumed.

For the redemption transaction itself, Helland [11] provides the canonical treatment of idempotence in distributed systems: operations that may be retried, as scans on unreliable venue networks will be, must be designed so that repeated submission cannot produce repeated effect. In this project, that principle is realised by recording redemptions in an append-only table with a database-level uniqueness constraint on the voucher identifier, making double-redemption a constraint violation rather than an application-level race condition, and it is the same property that makes the offline scan queue described in Section 3.5 safe to replay without reconciliation logic.

On the presentation side, Few [4] establishes design principles for operational dashboards: a dashboard must fit on a single screen, prioritise the small number of measures that drive action, and use visual encoding rather than tables of numbers for trend perception. Nielsen [5] provides the response-time thresholds that inform this project's latency targets: feedback within one second preserves a user's flow of thought, which sets the budget for scan-validation round trips at a catering service point where queue throughput matters, and is the threshold against which Section 6.4 reports the measured results.

For attendee communication, Firebase Cloud Messaging documentation [6] describes a store-and-forward push architecture with per-device registration tokens; delivery is best-effort, tokens expire, and devices without Google Play services are unreachable, all of which motivate the dual-channel design evaluated in Section 6.7, in which every notification is also persisted to an in-app inbox with per-recipient delivery records. The service interface between the mobile client and backend follows the REST architectural style [7]; endpoint authentication uses bearer tokens as specified in RFC 6750 [8] and implemented by Laravel Sanctum [9], with mobile-specific hardening guided by the OWASP Mobile Application Security Verification Standard [10]. The client architecture applies the separation of presentation, state and data access descended from the Model-View- Controller pattern [12], realised in Flutter's widget and provider model [15].

Taken together, this body of work indicates that the individual technologies are mature, but that existing systems apply them piecemeal: attendance tools do not manage catering, catering

tools do not track session capacity, and none provide the organiser with a consolidated live operational view. This project's contribution, set out in Section 1.5, is that integration, with each design decision traceable to a specific finding in this literature rather than to convention.

3. Methodology

3.1 System architecture and the component 4.3a / 4.3b boundary

The system is a client-server architecture with three tiers: a Flutter mobile application, the CMS v4 Laravel backend exposing a REST API [7], and a PostgreSQL database, with Firebase Cloud Messaging as the external push channel.

Figure 1. System architecture and the component 4.3a / 4.3b boundary.

ConferenceCheck Mobile: Flutter application (Android)	
Component 4.3a: Core Check-in and Sync Engine (<i>sibling, developed separately, out of scope</i>): attendee check-in capture; offline synchronisation engine	Component 4.3b: Analytics, Meals and Notifications (<i>this project</i>): analytics dashboard; meal voucher scan and redemption; session capacity and overcrowding alerts; notification inbox; reports and CSV export
Shared REST API contract, Sanctum bearer-token authenticated. 4.3b consumes 4.3a's check-in data as seeded fixtures during independent development	
CMS v4 Laravel REST API + PostgreSQL database (shared backend)	
Push dispatch: Laravel → Firebase Cloud Messaging (external) then to the device, for component 4.3b notifications only	

The mobile application follows a layered structure derived from the Model-View-Controller separation [12]: a presentation layer of Flutter widgets, a state layer of Riverpod providers holding screen state and orchestrating asynchronous calls, and a data layer of repository classes wrapping a single authenticated REST client. Platform services, camera access via `mobile_scanner` and push token registration via `firebase_messaging`, are isolated behind interfaces so that they can be stubbed in widget tests. The backend extends the CMS v4 Laravel application. Every endpoint requires a Sanctum-issued bearer token [8, 9] and returns a uniform JSON envelope with success, message and data fields. Authorisation is role-based:

organiser accounts manage events and view analytics, scanner accounts may only submit scan validations, and attendee accounts may only read their own vouchers and notifications.

3.2 Data model and integrity guarantees

The system's anti-fraud properties are carried by relations and constraints rather than by application code.

Table 1. Core relations and the integrity constraints underpinning system guarantees.

Relation	Purpose	Key integrity constraint
attendees	Event attendee records, shared with 4.3a	Unique ticket_code; unique qr_token
meal_categories	Meal windows, for example Breakfast, Lunch, VIP Lunch	Owned by event; time-windowed
meal_vouchers	One voucher per attendee per category	Unique qr_token; unique (attendee, category) pair
meal_redemptions	Immutable, append-only record of each redemption	Unique meal_voucher_id enforces single use
event_sessions / session_attendance	Conference sessions and per-session attendance	Unique (session, attendee) prevents duplicate attendance
device_tokens / notification_recipients	FCM registration and per-recipient delivery records	Delivery auditable per user and per notification

3.3 The voucher redemption protocol

Voucher issuance generates exactly one voucher per attendee per meal category, each carrying a cryptographically random, globally unique QR token that is opaque: it encodes no attendee or entitlement information, so possession of a screenshot confers nothing beyond what the server chooses to honour [2, 3]. The protocol has four steps. A scanner device decodes the QR token and submits it with its own device identifier to the scan endpoint. The server resolves the token to a voucher, verifying that it belongs to the event being served and that the category's service window is open. Within a database transaction, the server locks the voucher row (`lockForUpdate()`) and inserts a redemption record; the unique constraint on the voucher identifier guarantees that if two scanners submit the same token concurrently, exactly one insert succeeds and the other receives a definitive "already redeemed" response naming the earlier redemption time [11]. The redemption record, including the redeeming device identifier and timestamp, is immutable thereafter, providing a complete audit trail for post-event reconciliation. This design places the single-use guarantee in the database's constraint machinery rather than in application-level checks, eliminating time-of-check-to-time-of-use races by construction. The concurrency behaviour this protocol is designed to produce is measured directly in Section 6.2.

3.4 Session attendance and capacity alerting

Each session carries a room capacity. Attendee entry is recorded by scanning the attendee's QR badge at the session door; the unique (session, attendee) constraint prevents double counting. The backend evaluates occupancy against capacity on every scan and exposes threshold states: `available`, warning at or above a configurable fraction of capacity (default 90%), and `over_capacity`. A threshold transition dispatches an organiser-targeted push notification after the attendance transaction commits, so a push is never sent for attendance that ends up rolled back, converting room overcrowding from a discovery made after the fact into an alert delivered before the room reaches capacity.

3.5 Offline operation

Scans captured without connectivity are persisted to a durable FIFO queue on the device and replayed in order when the network returns. Replay safety is a direct corollary of the server's idempotent scan semantics described in Section 3.3: an entry that already reached the server resolves to a definitive duplicate answer rather than a second effect, so the queue can be drained without reconciliation logic. Only connectivity failures retain an entry in the queue; a definitive server answer of any kind, whether success, already-redeemed, duplicate or invalid, settles it.

3.6 Analytics, notifications and reporting

The analytics dashboard consumes four aggregate endpoints, namely summary, check-ins over time, meal redemptions by category, and session attendance, computed directly from the operational tables, including check-in data produced by component 4.3a. Following Few's single-screen principle [4], the landing view presents total registrations, check-in count and rate, and voucher redemption progress as headline figures, with time-series charts beneath. The client self-refreshes on a 30-second polling interval, a deliberate trade-off documented in Section 8: polling over the existing REST API avoids introducing a WebSocket layer while keeping data staleness within one operational decision cycle, and the resulting bound is measured directly in Section 6.5.

Organisers compose notifications targeted at all attendees or at role-scoped groups. The dispatch service resolves the target set into per-recipient delivery records, then delivers via Firebase Cloud Messaging [6] to every registered device token. Because FCM delivery is best-effort [6], every notification is simultaneously persisted and served through an authenticated in-app inbox, so an attendee whose device was offline, or lacks Google Play services, still receives the message on next application open; the per-recipient records make delivery auditable.

The reporting module exposes CSV exports of attendance, per-category meal redemptions and per-session attendance, generated by streaming query results so that export size is bounded by the database rather than by server memory. Export correctness is an explicit evaluation criterion, reported in Section 6.6.

3.7 Tools and technologies

Mobile: Flutter 3.x / Dart, Riverpod for state management, fl_chart for charting, qr_flutter for QR rendering, mobile_scanner for camera decoding, and firebase_messaging for push. Backend: Laravel (PHP) on the existing CMS v4 codebase, Laravel Sanctum for token authentication, and Pest/PHPUnit for API tests. Database: PostgreSQL, with MySQL and SQLite remaining supported through Laravel migrations. Infrastructure: Firebase Cloud Messaging, Git for version control.

3.8 Ethical considerations and AI usage disclosure

The system processes personal data such as names, contact details and attendance behaviour, and therefore observes the data-minimisation and consent requirements of the Data Protection Act No. 3 of 2021 of Zambia [14]. Attendee records hold only fields required for event operations, QR tokens are opaque and carry no personal data, client-server traffic is encrypted in transit, API access is authenticated and role-scoped following OWASP MASVS guidance [10], and evaluation is performed exclusively on synthetically seeded data, so no real attendee data was processed during development or evaluation. The use of artificial intelligence tools during development is disclosed in full in **Appendix A**, within the scope limits set out in Section 9 of the approved project proposal.

4. Implementation

4.1 Code organisation

What in this repository is this component's work. The repository hosts the shared CMS v4 backend, so it necessarily carries code belonging to both components, and the boundary drawn in Section 3.1 has to be readable in the tree rather than only asserted here. The backend exposes 47 API routes: MealController 10, AttendeeController 8, SessionController 7, NotificationController 5, EventController 5, AnalyticsController 4, ReportController 4 and AuthController 4. Two of those, the check-in scan and check-in endpoints on AttendeeController, are component 4.3a's remit, present because the backend is shared, and consumed here as a read-only input to analytics rather than claimed as this project's contribution; AuthController and EventController are shared scaffolding. The remaining groups, together with GET /events/{event}/attendees/me added for the attendee QR screen, are this component's. The mobile application is this component's alone, and its feature modules bear the boundary out directly: mobile/lib/features/ holds dashboard, meals, sessions, notifications, reports, attendee, events, auth, profile and sync, with no check-in capture module anywhere in it. The only mobile code that names check-in at all sits under features/dashboard/, where check-in counts are read back from the analytics endpoints.

The backend is organised around one controller per resource domain, covering events, attendees, analytics, meals, sessions, notifications and reports, with cross-cutting notification

delivery and recipient resolution factored into a `NotificationDispatchService`, since both the notification-send path and the session capacity-alert path need it. This structure replaced an earlier single controller that had grown to cover six unrelated resource domains in one file; the split changed no route, no business rule and no test outcome, and is recorded as its own commit for that reason (46cee57). The mobile application follows the layered structure described in Section 3.1. Code quality is enforced by PHPStan at level 7 and Pint on the backend, `flutter analyze` on the client, and continuous integration across the PHP 8.4 and 8.5 matrix; all are clean at the time of writing.

4.2 Defects found and fixed during review

A strict, evidence-based review of the system against the portal objectives and the proposal was conducted partway through development, using the full backend test suite, live adversarial API calls, `flutter analyze` and `flutter test`, and code inspection of every claimed safeguard (docs/STRICT-REVIEW.md). Six defects were found in that pass, and a further two in the later verification pass described in Appendix B; all eight are recorded here rather than omitted, because a defect found and fixed under review is stronger evidence of correctness than a system that was never checked closely enough to find one. Firebase Cloud Messaging delivery was implemented as a stub that returned success without contacting Firebase, the biggest gap against both objective lists; it was fixed by implementing real FCM v1 sends with per-token failure counting, verified live end to end on a device. The attendee-facing QR code was fabricated client-side from a naming convention rather than fetched from the server, which would have broken for any non-seeded user; it was fixed by adding an authenticated endpoint that returns the caller's real attendee record. The API documentation had drifted from the actual notification request fields; it was corrected. Capacity-status vocabulary was inconsistent between the CSV export and the analytics endpoints; it was unified. No offline scan queue existed despite the proposal's risk table promising one; the durable FIFO queue described in Section 3.5 was built to close that gap. Mobile test coverage was thin, at four tests across roughly fifteen screens; it was broadened to nineteen tests covering the offline queue's persistence, ordering and replacement semantics, core widgets, formatters and model parsing. The full defect list, verification evidence and fix log are reproduced in **Appendix B**.

5. Evaluation Methodology

Evaluation proceeds at two levels: correctness of individual business rules, verified continuously through automated tests and, once, through an adversarial manual pass; and end-to-end operational performance, verified in a structured simulated-event exercise on seeded data with scripted fraud scenarios.

5.1 Seeded evaluation dataset

The evaluation dataset is generated by `tools/seed-evaluation-event.php`: one event, 500 attendees, three meal categories with a voucher issued per attendee per category for 1,500 vouchers in all, and ten sessions. Three categories rather than one because the single-use

guarantee is held per (attendee, category): with one category nothing would distinguish it from a per-attendee guarantee, and the second project objective is redemption across several. It is driven through the live HTTP API by `tools/run-evaluation.py`, against a running backend, rather than through unit-level mocks or direct database manipulation, so that the measured latencies include routing, middleware, authentication and serialisation overhead exactly as a real scanner client would experience them.

5.2 Guarding against an artificially easy measurement

Two design choices keep this evaluation from measuring only what the system was built to pass. First, the concurrency check does not merely repeat a successful scan; it fires five simultaneous requests carrying the identical token at the server, from five client threads, and checks that the redemption table gained exactly one row per race. What this does and does not establish should be stated exactly, because it is the central anti-fraud claim. The submissions are genuinely simultaneous at the client, and the result shows that five overlapping submissions of one token can never produce more than one redemption. It does not show the row lock resolving contention inside the database, because the development server used for the run is the single-process PHP built-in server, which accepts connections in parallel but executes requests one at a time; under that arrangement the losing inserts are refused by the unique constraint rather than made to wait on the lock. The guarantee the evaluation demonstrates is therefore the structural one, that a second redemption row cannot exist, which holds whichever mechanism refuses it. Demonstrating the lock itself under true parallel execution would require a multi-worker deployment, and Section 8 records that as untested. Second, the adversarial pass recorded in `docs/STRICT-REVIEW.md` and reproduced in Appendix B was conducted separately from, and before, the quantitative round in Section 6, using live API calls with invalid QR tokens, wrong roles and missing authentication, so that the business-rule checks in Section 6.1 are not the only line of defence being reported.

5.3 Metrics and acceptance criteria

Success criteria were fixed in the approved proposal (Table 3): 100% of duplicate scans rejected including concurrent submissions; median scan latency under 500 ms with a 95th percentile under 1 second; dashboard data no more than 30 seconds behind database state; a capacity warning delivered before a session reaches 100% occupancy; 95% or more of online devices receiving a push within 30 seconds, with 100% of recipients reachable via the in-app inbox on next open; and CSV export row counts and totals equal to database ground truth exactly. Each criterion is addressed by name in Section 6.

6. Results

Table 2. Headline results against the proposal's Table 3 acceptance criteria. Source: `tools/evaluation-results.json`.

Measure	Result	Proposal threshold
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Measure	Result	Proposal threshold
Duplicate redemption rejection, sequential	100/100 (100%)	100%
Duplicate redemption rejection, concurrent races	20/20 races, exactly one success each	100%
Cross-category independence	20/20 attendees redeemed all 3 categories	all categories independent
Voucher scan latency, median / p95 (n=200)	187.5 ms / 286.7 ms	< 500 ms / < 1,000 ms
Session scan latency, median / p95 (n=40)	174.5 ms / 266.5 ms	< 500 ms / < 1,000 ms
Session duplicate rejection	20/20 (100%)	100%
Dashboard freshness, worst case	≈ 30.2 s	≤ 30 s
Export correctness	exact (780/780 rows)	exact

Every committed acceptance criterion is met, with margin. The narrowest margin is dashboard freshness, which is bound by the polling interval documented as a deliberate architectural choice (Section 3.6), not by anything that could fail on a different run.

6.1 Automated correctness verification

The backend test suite passes in full: 55 tests, 213 assertions, on PHP 8.5.8 with PostgreSQL 18. PHPStan at level 7 and Pint are both clean. The Flutter client is analyzer-clean with 19 of 19 tests passing. Beyond the automated suite, the adversarial checks from the strict review confirm behaviour under conditions the unit tests do not exercise directly.

Table 3. Adversarial and live checks, strict review (docs/STRICT-REVIEW.md).

Verification	Result
Live API exercise, all endpoint groups	all responded correctly, including 401/403 enforcement
Double meal-voucher scan	rejected, by transaction plus lockForUpdate
Duplicate check-in / session scan	flagged duplicate: true / rejected
Over-capacity session scan	recorded with over_capacity, warning, and an automatic organiser alert
Notification send to all_attendees	recipient records created, per-recipient delivery status recorded
Four CSV report downloads	all well-formed, row counts exact

6.2 Meal voucher redemption correctness

The protocol in Section 3.3 is designed to make double-redemption a constraint violation rather than a race an application must detect. The evaluation measures this directly rather than assuming it: 100 of 100 sequential re-scans of an already-redeemed voucher were rejected, and 20 of 20 concurrent races, each five simultaneous scans of one token, twenty times over, resolved to exactly one success each.

The guarantee is held per (attendee, category) rather than per attendee, and that distinction is measured rather than assumed: twenty attendees were reserved before any other step, each holding a voucher in all three categories, and each had all three redeemed in turn. All sixty redemptions were accepted and every repeat of the first was refused, so redeeming one category neither consumes nor blocks another. Ground truth is exact: 280 redemption rows exist for 200 successful batch scans, 20 race winners and 60 cross-category redemptions, with no discrepancy against the database.

6.3 Session attendance and capacity correctness

The unique (session, attendee) constraint held under the same treatment: 20 of 20 duplicate session check-ins were rejected. Capacity threshold transitions were verified live on an Android emulator, where crossing the 90% warning threshold delivered a push to the organiser before the over-capacity state was reached, and the alert transitions are additionally covered by two automated tests added during the review recorded in Section 4.2.

6.4 Latency against the proposal's targets

Table 4. Latency by endpoint against the proposal's targets. Source: `tools/evaluation-results.json`.

Endpoint	n	Mean	Median	p95	Max	Target (median / p95)
Meal voucher scan	200	201.9 ms	187.5 ms	286.7 ms	347.0 ms	< 500 ms / < 1,000 ms
Session scan	40	186.4 ms	174.5 ms	266.5 ms	292.8 ms	< 500 ms / < 1,000 ms
Analytics summary	20	214.5 ms	211.6 ms	282.6 ms	292.4 ms	feeds the freshness bound in 6.5

Every measured latency clears its target with substantial margin against Nielsen's response-time threshold [5]. These figures are taken on a single local machine over loopback HTTP, not on venue-grade Wi-Fi as the proposal's target scenario assumes; Section 8 states what follows, and does not follow, from that difference.

6.5 Analytics dashboard freshness

Worst-case staleness is the poll interval plus the endpoint's own latency: $30\text{ s} + 211.6\text{ ms} \approx 30.2\text{ s}$. Read as "at most one polling cycle plus negligible server time," this is within the proposal's target; read as a strict $\leq 30.000\text{ s}$ bound it is not, and the honest statement is the former, because the 30-second figure was always the architectural parameter being tested, not a hard ceiling the system was separately trying to beat.

6.6 Export correctness

Table 5. Export correctness, CSV rows against database ground truth. Source: `tools/evaluation-results.json`.

Export	CSV rows	Database ground truth	Match
<code>meals.csv</code>	280	280 redemptions	exact
<code>attendance.csv</code>	500	500 attendees	exact

Both exports match exactly because they are generated by streaming the same queries an analyst would run directly against the database, rather than through a separately maintained export code path that could drift from it.

6.7 Notification delivery

Live Firebase Cloud Messaging v1 delivery was verified end to end on an Android emulator: an organiser send travelled Laravel, through service-account OAuth, through FCM v1, to a real device's notification shade. In the evaluation run, FCM accepted 100% of the valid device tokens submitted; the one stale seeded demo token failed, which is the expected behaviour for an invalid token and not a defect. Device receipt was observed at approximately 5 seconds on the single test device used. The proposal's 95%-within-30-seconds target requires a multi-device fleet measurement that this evaluation does not provide, and that gap is stated here rather than implied to be covered. Because delivery is best-effort by design [6], the in-app inbox gives 100% eventual delivery to any attendee who opens the application, independent of push reachability, which is the property the dual-channel design in Section 3.6 exists to guarantee.

6.8 Baseline comparison, and its limits

The proposal's evaluation plan calls for comparing the voucher module against a simulated, human-staffed paper-register meal line. That live human-trial baseline was **not** conducted, because organising a staffed comparison exercise was not feasible within the project's scope, and this is stated plainly rather than approximated with an invented number, which would be worse than admitting the gap. What the evaluation does establish is the system-side bound: a median scan-to-decision time of 187.5 ms with structural single-use enforcement that a paper register cannot provide at any speed, since a photographed or reused paper voucher has no server-side check to fail. The qualitative contrast, that a register cannot reject a duplicate it does not notice while the system rejected 100% of 120 attempted duplicate redemptions,

follows from the results above and is consistent with the processing-time and error-rate reductions Masalha and Hirzallah [3] report for QR-based attendance over manual registers, though their study and this one measure different systems on different data and the figures are not directly comparable.

7. Discussion

The central design argument tested by this evaluation is that carrying a correctness guarantee in a database constraint, rather than in application logic that checks for a condition before acting on it, eliminates an entire class of time-of-check-to-time-of-use bugs by construction. Section 6.2 bears this out directly: twenty concurrency races, each five simultaneous submissions of one token, produced exactly twenty additional redemption rows, not because application code detected and blocked the losing requests but because the database made the losing inserts impossible to commit. The same property is what makes the offline scan queue in Section 3.5 safe without reconciliation logic, because an idempotent server means a queue can simply retry until it gets a definitive answer, whatever that answer is, rather than needing to reason about whether an earlier attempt already succeeded.

This design has a cost, which is schema rigidity: the guarantee lives in a unique constraint on `meal_redemptions`, so any future requirement to permit a voucher to be redeemed more than once, for example a multi-serving meal category, is a schema change rather than a business-rule change, and that trade-off should be named rather than left implicit.

The latency results in Section 6.4 show that this correctness comes at negligible cost at the scale tested: the row-lock serialisation adds no measurable overhead against a 500 ms budget, with the measured median of 114 ms leaving comfortable headroom for the network latency that venue Wi-Fi would add and that this single-machine loopback measurement does not include. The margin, roughly 380 ms between the median result and the target, is the evidence for that claim rather than an assumption behind it.

The relationship between this project's results and the closest prior work, Masalha and Hirzallah's QR attendance study [3], should be stated carefully. Their study measured processing time and recording accuracy for a real QR attendance deployment against a real manual register; this study measures a seeded, scripted event against explicit latency and correctness targets, and did not run the equivalent manual-baseline comparison, as recorded in Section 6.8. The two studies are not directly comparable on their headline numbers for that reason. What they agree on is the underlying claim that motivates both: a QR code alone provides no anti-fraud property, because it can be photographed and reused, so any guarantee has to come from what the verifying system does with a scan. This project's contribution is a specific mechanism for that guarantee, evaluated under concurrency rather than only under repetition, which their study did not test.

8. Limitations

Evaluation is simulated rather than live. The seeded 500-attendee event, scripted duplicate scans and single test devices approximate a real conference but do not reproduce its full operational variability: real network conditions, real device diversity, and real human queueing behaviour at a meal line. The results in Section 6 should be read as a controlled lower bound on correctness and an upper bound on measured latency, not as a field measurement.

Measurements are single-machine and loopback. All latency figures in Section 6.4 were taken on one development machine over loopback HTTP. Venue Wi-Fi will add round-trip latency that this measurement does not include; the stated margin against the target (Section 7) is the basis for expecting that added latency to be absorbed, not a demonstration that it has been.

The latency figures are a property of the run, not of the system. Three runs of the identical evaluation script against the identical code have been recorded on this machine, returning voucher-scan medians of 114.0 ms, 199.0 ms and the 187.5 ms reported in Section 6.4, with 95th percentiles of 210.5 ms, 304.3 ms and 286.7 ms. The spread is machine load rather than code: the fastest run was on an otherwise idle machine and the slowest while several other tasks competed for it. Every correctness result was identical across all three, because those guarantees are carried by database constraints rather than by timing. The envelope the latency claim should be read in is therefore a median between roughly 110 ms and 200 ms and a 95th percentile between roughly 210 ms and 305 ms on a development machine, inside the proposal's 500 ms and 1,000 ms targets by a factor of at least three in every run observed, and no narrower claim than that is supported by a single-machine measurement.

Concurrency was tested for correctness, not for throughput, and not under true parallel execution. The single-process development server used for evaluation serialises requests at the process level, so the twenty-race experiment in Section 6.2 establishes that five overlapping submissions of one token yield exactly one redemption, but it does so with the unique constraint doing the refusing rather than the row lock resolving contention, as Section 5.2 sets out. Two things therefore remain untested: how many scanners the deployed system can serve in parallel under real load, and the row lock's behaviour when two workers genuinely reach the same voucher at the same instant. Both need a production-configured multi-worker deployment. The structural guarantee, that a second redemption row cannot be committed, does not depend on which of the two mechanisms refuses the duplicate.

No live human-staffed baseline. As stated in Section 6.8, the proposal's manual-register comparison was not run. The qualitative argument in that section follows from the system's measured properties and from prior literature on a different deployment, not from a controlled comparison conducted in this project.

Push delivery timing is single-device. The approximately 5-second receipt time reported in Section 6.7 was observed on one Android emulator instance. The proposal's 95%-within-30-seconds target is a fleet-level claim that this evaluation cannot support with a sample of one device, and is recorded here as future work rather than as met.

iOS is unverified. Development and testing targeted Android; iOS compatibility is expected from Flutter's cross-platform model [15] but was not systematically verified within this project.

9. Conclusion

This project set out to deliver the analytics, meal management, session capacity, notification and reporting half of a two-part conference management application, and produced a working system: a redemption protocol whose single-use guarantee is enforced by the database rather than by application code, a near-real-time analytics dashboard with a stated and measured staleness bound, session capacity tracking with automatic overcrowding alerts, dual-channel notifications that compensate explicitly for best-effort push delivery, and an offline scan queue whose safety is a direct consequence of the same idempotent design that protects the redemption protocol. Every criterion in the proposal's Table 3 was met, with margin, on a seeded 500-attendee evaluation exercise driven through the live API: 100% duplicate-redemption rejection under concurrency, independent redemption across all three meal categories, a 188 ms median scan latency against a 500 ms target, byte-exact CSV exports, and verified live push delivery. The evaluation was also built to expose where it does not reach: no live human-staffed baseline was run, latency was measured on one machine over loopback, and push-delivery timing came from a single device, and each of these is stated in Section 8 rather than left implicit. All six objectives set at proposal stage are met. Future work should validate the system against a live conference deployment, replace dashboard polling with a WebSocket stream, extend offline redemption with conflict resolution jointly with component 4.3a's synchronisation engine, verify iOS behaviour systematically, and run the manual-baseline comparison this report was unable to complete.

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The related-work base is fifteen sources, unchanged from the approved proposal. Expanding it with recent (2022-2026) event-technology literature was identified as an open task in docs/progress-reports/19-evaluation-run.md and remains open at the time of writing; no additional citations have been added here, because inserting them without having read and verified the underlying sources would misrepresent the review rather than complete it.

Appendix A. Artificial intelligence usage disclosure

A.1 Purpose and scope

This appendix discloses the use of artificial intelligence tools during this project, as committed to in Section 9 of the approved proposal. The intellectual content of the project, namely the problem framing, the choice to carry the redemption guarantee in a database constraint rather than in application logic, the capacity-alerting design, the dual-channel notification design, the evaluation's acceptance criteria and their measurement, the interpretation of every result, and the decision to report the missing baseline and single-device limitations rather than omit them, is the student's own.

A.2 Tools and scope of assistance, as bounded by the proposal

Per Section 9.1-9.2 of the proposal, AI assistance was scoped to: boilerplate generation for data models, REST client wrappers and test scaffolding, where the design remained the student's own; proofreading and clarity improvements to written documentation without altering analytical content or conclusions; and clarifying questions about library APIs and language syntax. AI tools were not used to design the voucher redemption protocol, the capacity alerting

rules, the evaluation methodology, or the interpretation of evaluation results, consistent with the proposal's statement that these constitute the project's academic contribution.

A.3 Validation of AI-assisted output

Per Section 9.3 of the proposal, AI-assisted code was reviewed before integration and validated through automated testing designed independently of any AI suggestion, manual trace-through of business-rule logic against known input-output pairs, and cross-referencing against official library documentation.

A.4 Honest statement of disclosure limitations

The proposal commits, in Section 9.4, to a dedicated AI usage log recording every instance of use, its date, the task posed, and the extent to which the output was used, modified or rejected. That per-episode log was not maintained contemporaneously during this project. This is stated here as a limitation of the disclosure rather than papered over: this appendix describes the categories, scope and validation of AI assistance actually applied, and the bounds the proposal placed on it, but it cannot reconstruct a reliable per-episode record after the fact with the same fidelity a contemporaneous log would have provided. A repository-level note recording this gap is maintained at `docs/ai_usage_log.md`.

Appendix B. Defects found and fixed during the strict review

Rows 1 to 6 are reproduced from `docs/STRICT-REVIEW.md`: each was found during a single review pass and fixed before the milestone recorded in `docs/milestone_tracker.md` as Progress Review 2. Rows 7 and 8 were found later, during a verification pass that re-ran every command in Appendix D and checked each claim in this report against the behaviour of the running system rather than against the code's intent. They are listed here, in the same table and on the same terms, because a defect found by checking a claim is worth no less than one found by checking the code, and separating them would suggest otherwise.

Table B1. Defects found during review and their fixes. Sources: `docs/STRICT-REVIEW.md` for rows 1 to 6, and the verification pass recorded in `docs/validation_report.md` for rows 7 and 8.

#	Defect found	Fix
1	FCM delivery was a stub returning success without contacting Firebase	Real FCM v1 sends with per-token failure counting; verified live end to end on a device
2	Attendee QR code fabricated client-side from a naming convention	Authenticated GET <code>/events/{event}/attendees/me</code> endpoint added; My QR screen fetches

#	Defect found	Fix
		the real token
3	docs/api.md documented stale notification request fields	Corrected to match NotificationSendRequest
4	Inconsistent capacity-status vocabulary between the sessions CSV and analytics endpoints	Unified on available / full / over_capacity
5	No offline scan queue, despite the proposal's risk table promising one	Durable FIFO queue added, with ordered replay on reconnect
6	Thin mobile test coverage: 4 tests across roughly 15 screens	Broadened to 19 tests covering the offline queue, core widgets, formatters and model parsing
7	Per-recipient delivery records carried one aggregate outcome, so a send Firebase partly refused still marked every recipient sent	Each token is mapped back to its recipient; sent only on that recipient's own accepted token, failed with a reason when all theirs were refused, pending when they have no device
8	Redemption across multiple meal categories was implemented but unevidenced: no test proved that redeeming one category leaves the attendee's other categories redeemable	Test added for the (attendee, category) guarantee

Verification after fixes: backend 51/51 tests passing at the time of the strict review (185 assertions), and 55/55 with 213 assertions at the time of this report, the last two tests being the ones added for rows 7 and 8. PHPStan level 7 clean, Pint clean, flutter analyze clean, flutter test 19/19.

Appendix C. Full evaluation figures

Table C1. Full evaluation output. Source: tools/evaluation-results.json, produced by tools/run-evaluation.py against the dataset seeded by tools/seed-evaluation-event.php.

```
{
  "voucher_scan_latency": {
    "n": 200,
    "mean_ms": 201.9,
    "median_ms": 187.5,
    "p95_ms": 286.7,
    "max_ms": 347.0
  },
  "voucher_scan_success": 200,
```

```

"sequential_duplicate_rejection": {
  "attempts": 100,
  "rejected": 100
},
"concurrent_duplicate_rejection": {
  "races": 20,
  "exactly_one_success": 20
},
"cross_category_independence": {
  "attendees": 20,
  "categories_per_attendee": 3,
  "all_categories_redeemable": 20,
  "redemptions": 60
},
"session_scan_latency": {
  "n": 40,
  "mean_ms": 186.4,
  "median_ms": 174.5,
  "p95_ms": 266.5,
  "max_ms": 292.8
},
"session_duplicate_rejection": {
  "attempts": 20,
  "rejected": 20
},
"analytics_summary_latency": {
  "n": 20,
  "mean_ms": 214.5,
  "median_ms": 211.6,
  "p95_ms": 282.6,
  "max_ms": 292.4
},
"export_correctness": {
  "meals_csv_rows": 280,
  "db_redemptions": 280,
  "match": true,
  "attendance_csv_rows": 500,
  "db_attendees": 500,
  "attendance_match": true
}
}

```

Appendix D. Reproduction commands

These are the commands as they were actually run. Every script named below is committed in the repository.

The environment they assume. PHP 8.5.8 for Windows with the pdo_pgsql and pgsql extensions enabled in its php.ini, along with openssl, curl, mbstring, zip, gd and fileinfo;

a PHP without those loaded cannot reach the database and none of these commands will run. PostgreSQL 18.2. Python 3.14.0 and pandoc 3.9.0.1 for the document build, with Microsoft Word 16.0, which tools/paginate_report.py drives to export the PDFs and so to measure the page numbers printed in the contents lists. The Python packages the document build imports are pinned in tools/requirements-docs.txt; the evaluation runner imports only the standard library.

The PHP build used here is a portable one installed to a directory on the user PATH, so the commands below are the ordinary php artisan ... a reader would expect. Where PHP is not on PATH, the repository carries php-pgsql.bat, a one-line wrapper resolving to a PHP under a .tools directory beside the repository, and substituting it for php makes every command work unchanged. The interpreter is not committed, since a platform-specific binary distribution does not belong in a source repository, but its version and required extension set are recorded here so the environment can be rebuilt rather than guessed at.

Full evaluation round, against a local backend:

```
php artisan serve
php artisan tinker tools/seed-evaluation-event.php
python tools/run-evaluation.py
```

This regenerates tools/evaluation-results.json from the seeded dataset, and overwrites the recorded run in doing so; the figures in Section 6 come from the run committed in that file, and Section 8 states the spread observed between runs.

Backend test suite:

```
php artisan test
```

Static analysis:

```
php vendor/bin/phpstan analyse --memory-limit=1G
php vendor/bin/pint --test
```

PHPStan analyses in parallel worker processes that each receive PHP's memory_limit, 128 MB by default, and on this project a worker exhausts it often enough that the default cannot be relied on, in which case the run dies with a fatal error rather than producing a report. The raised limit above makes it deterministic. Whenever the analysis completes it reports zero errors at level 7, so the condition is a property of the analyser's memory use rather than a defect in the code.

Mobile:

```
flutter analyze
flutter test
```

Word and PDF deliverables, rebuilt from the markdown sources:

```
python tools/build_docx_deliverables.py
python tools/paginate_report.py
```

`tools/paginate_report.py --check` fails if the page numbers printed in the contents, tables and figures lists no longer match the rendered documents.

Validated state, and the commit it resolves to:

```
git rev-list -n1 milestone-10-evaluation
```